

# Chapter 1

## Exam Reviews

### 1.3 Review for Exam 3

#### 1.3.1 Wilson's Theorem

**Theorem 1 (Wilson's Theorem)**

If  $p$  is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

**Theorem 2 (Some Unnamed Theorem)**

If  $n$  is a composite integer greater than 4,

then

$$(n-1)! \equiv 0 \pmod{n}$$

If  $n = 4$ , then

$$3! \equiv 2 \pmod{4}$$

### 1.3.2 Fermat's Little Theorem

#### Theorem 3 (Fermat's Little Theorem)

If  $a \in \mathbb{Z}$  with,

1.  $p$  is prime
2.  $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

### 1.3.3 Extended Fermat's Little Theorem

#### Theorem 4 (Extended Fermat's Little Theorem)

If  $a, p, q \in \mathbb{Z}$  with

1.  $p$  and  $q$  are distinct primes
2.  $\gcd(a, pq) = 1$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

### 1.3.4 Bezout's Theorem

#### Theorem 5 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

When given a problem that's like find  $x, y$  in  $ax + by = \gcd(a, b)$  (where they give you  $a$  and  $b$ ), just do Euclidean and stop when you get to the actual GCD then back solve

### 1.3.5 Is a number prime or composite?

Lowkey just brute force up until

$$\lceil \sqrt{n} \rceil$$

### 1.3.6 Prime Factorization

Let's look at how to find the number of divisors in a number  $n$

#### Definition 1 (Prime Factorization)

**Prime factorization** is breaking up a number into prime factors. An example is

$$\begin{aligned} 24 &= 6 \cdot 4 \\ &= (3 \cdot 2)(2 \cdot 2) \\ &= 3^1 \cdot 2^3 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up  $n$  into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

$p_k$ : Prime factor  
 $e_k$ : Occurences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$

### 1.3.7 Chinese Remainder Theorem

Given the system: (I'm just gonna use up to 3 for this)

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ x &\equiv a_3 \pmod{m_3} \end{aligned}$$

The system has a **unique solution** if  $m_1, m_2, m_3$  are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1 k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$\begin{aligned} x &\equiv a_2 \pmod{m_2} \\ m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \end{aligned}$$

And solve for  $k_1$

$$\begin{aligned} m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\ m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\ m_1^{-1} k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ \boxed{k_1 \equiv c_1} &\pmod{m_2} \quad (c_1 \text{ is shorthand}) \end{aligned}$$

**Note**

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of  $m_1$

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, k_2 \in \mathbb{Z}$$

3. Plug that into our equation for  $x$

$$x = m_1 k_1 + a_1$$

$$\boxed{x = m_1(m_2 k_2 + c_1) + a_1}$$

Plug that into the last equation

$$m_1(m_2 k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for  $k_2 \pmod{m_3}$

$$\vdots$$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3 k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for  $x$

$$x = m_1(m_2 k_2 + c_1) + a_1$$

$$x = m_1(m_2(m_3 k_3 + c_2) + c_1) + a_1$$

$$\boxed{x = m_1 m_2 m_3 k_3 + c_3}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3 \pmod{m_1 m_2 m_3}}$$

Basically, starting with the first equation

1. Expand first, solving for  $x$
2. Plug  $x$  into second
3. Solve for  $k_1$
4. Expand  $k_1$
5. Plug  $k_1$  back into first expanded of  $x$  (one with  $k_1$ )
6. Solve for  $k_2$
7. Expand  $k_2$
8. Plug  $k_2$  back into second expanded of  $x$  (one with  $k_2$ )
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation

## 1.4 Exam 4 Review

### 1.4.1 Binary and Base $n$

Write 1577 in binary

$$\begin{aligned}
 1577 &= 2 \cdot 788 + 1 \\
 788 &= 2 \cdot 394 + 0 \\
 394 &= 2 \cdot 197 + 0 \\
 197 &= 2 \cdot 98 + 1 \\
 98 &= 2 \cdot 49 + 0 \\
 49 &= 2 \cdot 24 + 1 \\
 24 &= 2 \cdot 12 + 0 \\
 12 &= 2 \cdot 6 + 0 \\
 6 &= 2 \cdot 3 + 0 \\
 3 &= 2 \cdot 1 + 1 \\
 1 &= 2 \cdot 0 + 1
 \end{aligned}$$

Writing the bottom to top left to right...

$$\boxed{11000101001}$$

We could check this by just doing

$$1577 = 2^{10} + 2^9 + \dots$$

Let's try representing 1577 in base 8. Remember that each remainder we get is the coefficient to the corresponding power

$$1577 = 8 \cdot 197 + \mathbf{1}$$

$$197 = 8 \cdot 24 + \mathbf{5}$$

$$24 = 8 \cdot 3 + \mathbf{0}$$

$$3 = 8 \cdot 0 + \mathbf{3}$$

Rewriting this, we get

$$\underbrace{3}_{8^3} \underbrace{0}_{8^3} \underbrace{5}_{8^2} \underbrace{1}_{8^0}$$

So

$$1577 = 3 \cdot 8^3 + 5 \cdot 8^1 + 1 \cdot 8^0$$

Now let's do it with base 16

$$1577 = 16 \cdot 98 + \mathbf{9}$$

$$98 = 16 \cdot 6 + \mathbf{2}$$

$$6 = 16 \cdot 0 + \mathbf{6}$$

$$\boxed{1577 = 629_{16}}$$

$$1577 = 6 \cdot 16^2 + 2 \cdot 16^1 + 9 \cdot 16^0$$

For our last example, write the  $AB0CD_{16}$  in base 10 (aka normal numbers)

Recall that

$$A = 10$$

$$B = 11$$

$$\vdots$$

$AB0CD$  is just 10 11 0 12 13

Since there's 5 "numbers" that means we start with  $16^{5-1} = 16^4$

$$\begin{aligned} AB0CD_{16} &= 10 \cdot 16^4 + 11 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 13 \cdot 16^0 \\ &= \boxed{700621} \quad (\text{just plug into calc}) \end{aligned}$$